



S V V SAI SRI VALLABH PATANENI

Fuel Cell System Engineer · Hydrogen & Fuel Cell R&D

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PROFILE

My path into fuel cell systems began hands-on at Ecogenium e.V., where I fabricated a standalone test bench, ran an air-cooled PEM stack through its operating window, and wrote the energy-management and thermal-control strategies that carried Germany's only student-built hydrogen car to a runner-up finish at Shell Eco-Marathon 2025. That experimental grounding took me to Robert Bosch GmbH at system level, engineering FCEV operating strategies and studying fuel-cell degradation, and then to DLR, where I am now developing a complete ~150 kW fuel cell system for a hydrogen aircraft, from a thermal-management digital twin and component sizing to the control software itself. I want to bring this full-stack experience, from stack to system to control, to the hydrogen and fuel cell research at Forschungszentrum Juelich.

PROFESSIONAL EXPERIENCE

- Werkstudent, Fuel Cell Systems (Aircraft)**
DLR, Innovation Center for Small Aircraft Technologies (INK)

01/2026 – Present
Wuersele (Aachen), Germany

 - Fuel Cell System Development:** Developing the complete ~150 kW fuel cell system (the engine) for D-LIGHT, DLR's hydrogen-powered 9-seater feeder aircraft, from concept toward a flight-representative system.
 - Thermal-Management Digital Twin:** Built a MATLAB/Simulink digital twin of the full thermal management system to derive design specifications (volumetric flow, coolant temperature range, coolant pressure), driving procurement of the coolant pump and heat exchanger.
 - Sizing & Component Selection:** Selected all thermal-management components against the derived specifications and sized the subsystem for the aircraft duty cycle.
 - System Rig Design:** Iteratively designed a compact, aircraft-engine-sized fuel cell system rig in CATIA V5, optimised for easy transferability and integration.
 - Upcoming Master's Thesis:** Developing the fuel cell system control (start-up, purge control, cathode and anode subsystem control, temperature control, safety protocols) in MATLAB and C++ for direct deployment to the fuel cell controller.
- Operating Strategy / System Simulation Intern**
Robert Bosch GmbH

09/2025 – 12/2025
Schwieberdingen, Germany

 - Energy-Management & Degradation:** Engineered a MATLAB operating strategy for Strong Fuel Cell Hybrid EVs that safeguards fuel-cell and HV-battery degradation by managing dynamic loads, thermal stress, low-voltage operation, SOC swing and life cycles, parameterised via MAPs for different battery capacities and drive cycles.
 - Model Development & Validation:** Developed and validated an iterative Simulink power-split operating-strategy model in the (FC)EVsim module, correlating outputs against reference measurement data to ensure high model fidelity.
 - Optimisation & Benchmarking:** Executed multi-objective parametric optimisation (Genetic Algorithms, Gradient Descent) to generate control MAPs for charge sustenance and minimal fuel consumption, benchmarked against DP and ECMS; co-authoring a peer-reviewed paper on the strategy.
- Fuel Cell Systems & Test-Bench Lead (Team Lead)**
Ecogenium e.V (RWTH student team), Shell Eco-Marathon

06/2024 – Present
Aachen, Germany

 - Air-Cooled Fuel Cell Test Bench:** Engineered, fabricated and electronically integrated a standalone fuel cell test bench; ran purging, humidity cycling, short-circuiting and polarisation-curve experiments and mapped the stack's optimal thermal operating window.
 - Energy-Management & Thermal Control:** Architected and deployed energy-management and fuel-cell temperature-control strategies in Python/Simulink (MIL-validated), boosting on-track fuel efficiency by ~80% and securing a runner-up finish at Shell Eco-Marathon 2025.
 - System Integration:** Designed and tested a compact fuel cell system rig, optimising packaging density and integration of the fuel cell and its electronic systems for Germany's only student-built hydrogen fuel cell vehicle.

KEY SKILLS

FUEL CELL SYSTEMS	PEM fuel cell systems · air-cooled stack operation · balance of plant · thermal management · system sizing · component selection · FCEV powertrain
CONTROLS & STRATEGY	operating strategy · energy management (EMS) · degradation / SOH analysis · MAP / lookup-table optimisation · DP / ECMS benchmarking
TEST & EXPERIMENTATION	FC test-bench design · polarisation curves · purging / humidity cycling · data acquisition · model validation
SIMULATION & MODELLING	MATLAB/Simulink (MIL) · digital twin · ANSYS · Star-CCM+ · Dymola
PROGRAMMING & DATA	Python (NumPy, Pandas, scikit-learn) · C/C++ · Power BI
CAD	CATIA V5 · Siemens NX · Fusion 360

EDUCATION

- M.Sc. Automotive Engineering**
RWTH Aachen University

09/2023 – 06/2026
GPA 1.7 (1.2 core courses)

Modules: Fuel Cell System Technology, Alternative & Mobile Propulsion Systems, Thermodynamics, Simulation Sciences, FEM/CFD. Thesis: fuel cell system control (MATLAB/C++).
- B.Tech Mechanical Engineering**
Jawaharlal Nehru Technological University, Hyderabad

06/2019 – 04/2023
GPA 1.3 (German scale)

SELECTED PROJECTS & PUBLICATIONS

- Co-author (in preparation): rule-based FCEV powertrain control strategy, based on the Bosch internship.
- Electric Wheelchair with Modular Battery System (ISEA-RWTH): battery-pack architecture, cell selection, Simulink discharge/range model.
- Paper: A Comprehensive Analysis of Two Innovative Aircraft Design Configurations (IJSED).

LANGUAGES

English (C1, fluent) · German (B1, intermediate) · Telugu / Hindi (native)